Tony Menzo

Theory Division, Fermi National Accelerator Laboratory
P.O. Box 500, MS 106, Batavia, IL 60510-5011, USA
Email: amenzo@ua.edu
Website: https://tonymenzo.com
INSPIRE: https://inspirehep.net/authors/2049582

Academic Appointment

Sep. 2025–present **Postdoctoral Fellow, Particle Theory and Phenomenology**
Joint appointment: Fermi National Accelerator Laboratory and the University of Alabama

Research Interests

Particle phenomenology: rare lepton decays, dark sectors, and effective field theory; analytic, effective, stochastic, and machine-learning-based models of hadronization; optimal transport as a measure of CP violation; reliable systems for autonomous agentic phenomenology, including deployment and evaluation on high-performance computing platforms.

Education

2020–2025 **Ph.D. in Physics**, University of Cincinnati
Advisor: Jure Zupan
Dissertation: *Rare Lepton Decays and Differentiable Hadronization Models: From Signatures of New Physics to Data-driven Event Generation*

2014–2019 **Bachelor of Arts and Sciences in Physics**, The Ohio State University
Minor: Earth Science (Geophysics)
Research mentor: Stuart Raby

Fellowships and Awards

2024–2025 **URA Visiting Scholars Program**, Fermilab
Time-dependent signals at the intensity frontier; one-month visit

2024 **URA Visiting Scholars Program**, Fermilab
Exploring exotic signatures at Mu2e; three-month visit

2023–2024 **DOE Office of Science Graduate Student Research (SCGSR)**
Lawrence Berkeley National Laboratory; six-month research placement

2023 **First place**, University of Cincinnati Physics Poster Competition

2022 **Sandwich Scholarship**, Technion–Israel Institute of Technology
Three-month visiting research studentship

Academic Service

Referee, Journal of High Energy Physics (JHEP); PRX Intelligence.

Co-organizer, Machine Learning Journal Club, University of Cincinnati, 2024–present.

Organizing committee member, Agentic AI for BSM Event Generation and Analysis Workshop at the LPC, Fermilab, September 9–11, 2026.
https://indico.fnal.gov/event/75639/

Co-organizer, University of Alabama HEP Seminar, 2025–present.

Publications and Preprints

1. George T. Fleming, Daniel C. Hackett, James Halverson, T. J. Hobbs, Stefan Höche, Brandon Kriesten, Konstantin T. Matchev, **Tony Menzo**, Stephen Mrenna, Benjamin Nachman, Prasanth Shyamsundar, David Shih, Jesse Thaler, Jure Zupan, et al. "Blueprint for an AI-Enabled Particle Theory Community." Preprint. arXiv:2610.28505 (2026).

2. Benoit Assi, Christian Bierlich, Rikab Gambhir, Philip Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Jure Zupan. "Pythia-Contrib: a modular contribution framework for the Pythia 8 event generator." Preprint. arXiv:2610.19111 (2026).

3. **Tony Menzo**, Alexander Roman, George T. Fleming, Sergei Gleyzer, Konstantin Matchev, Stephen Mrenna. "HEPTABench: A framework for portable and controlled agentic evaluations in high energy physics." Preprint. arXiv:2610.18454 (2026).

4. **Tony Menzo**, George T. Fleming, Konstantin T. Matchev, Stephen Mrenna, Alexander Roman. "Reining in an Agentic Harness for High Energy Physics." Preprint. arXiv:2609.00107 (2026).

5. Benoit Assi, Christian Bierlich, Rikab Gambhir, Philip Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Ahmed Youssef, Jure Zupan. "Vistas: A Visualization Interface for Particle Collision Simulations." Preprint. arXiv:2606.19524 (2026).

6. Yuval Grossman, **Tony Menzo**, Stefan Schacht, Chinhsan Sieng, Jure Zupan. "Unbinned extraction of γ from $B \to DK$ with normalizing flows." Preprint. arXiv:2605.06768 (2026).

7. Yohei Ema, Patrick J. Fox, Matheus Hostert, **Tony Menzo**, Maxim Pospelov, Anupam Ray, Jure Zupan. "New physics in multi-lepton tau decays." Preprint. arXiv:2604.01465 (2026).

8. **Tony Menzo**, Alexander Roman, George T. Fleming, Sergei Gleyzer, Konstantin T. Matchev, Stephen Mrenna. "Agentic Diagrammatica: Towards Autonomous Symbolic Computation in High Energy Physics." Preprint. arXiv:2603.26990 (2026).

9. **Tony Menzo**. "Conservation laws and effective hadronization models." Preprint. arXiv:2602.12599 (2026).

10. Benoît Assi, Christian Bierlich, Rikab Gambhir, Phil Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Jure Zupan. "HDSense: An efficient method for ranking observable sensitivity." Preprint. arXiv:2602.01509 (2026).

11. **Tony Menzo**, Alexander Roman, Sergei Gleyzer, Konstantin Matchev, George T. Fleming, Stefan Höche, Stephen Mrenna, Prasanth Shyamsundar. "HEPTAPOD: Orchestrating High Energy Physics Workflows Towards Autonomous Agency." Preprint. arXiv:2512.15867 (2025).

12. Anja Butter, Ayodele Ore, Sofia Palacios Schweitzer, Tilman Plehn, Benoît Assi, Christian Bierlich, Philip Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Ahmed Youssef, Jure Zupan. "Iterative HOMER with uncertainties." SciPost Phys. 20, 042 (2026). arXiv:2509.03592 (2025).

13. Yohei Ema, Patrick J. Fox, Matheus Hostert, **Tony Menzo**, Maxim Pospelov, Anupam Ray, Jure Zupan. "Long-lived Axion-Like Particles from Tau Decays." Phys. Rev. D 112, 115028 (2025). arXiv:2507.15271 (2025).

14. Benoît Assi, Christian Bierlich, Philip Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Ahmed Youssef, Jure Zupan. "Post-hoc reweighting of hadron production in the Lund string model." SciPost Phys. 19, 104 (2025). arXiv:2505.00142 (2025).

Publications and Preprints (continued)

15. Innes Bigaran, Patrick J. Fox, Yann Gouttenoire, Roni Harnik, Gordan Krnjaic, **Tony Menzo**, Jure Zupan. "Direct Detection of Ultralight Dark Matter via Charged Lepton Flavor Violation." Phys. Rev. Lett. 136, 251801 (2026). arXiv:2503.07722 (2025).

16. Benoit Assi, Christian Bierlich, Philip Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Ahmed Youssef, Jure Zupan. "Characterizing the hadronization of parton showers using the HOMER method." SciPost Phys. 19, 125 (2025). arXiv:2503.05667 (2025).

17. Chaja Baruch, Patrick J. Fitzpatrick, **Tony Menzo**, Yotam Soreq, Sokratis Trifinopoulos, Jure Zupan. "Searching for exotic scalars at fusion reactors." JHEP 10 (2025) 215. arXiv:2502.12314 (2025).

18. Nick Heller, Phil Ilten, **Tony Menzo**, Stephen Mrenna, Benjamin Nachman, Andrzej Siodmok, Manuel Szewc, Ahmed Youssef. "Rejection Sampling with Autodifferentiation - Case study: Fitting a Hadronization Model." Preprint. arXiv:2411.02194 (2024).

19. Christian Bierlich, Phil Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Ahmed Youssef, Jure Zupan. "Describing Hadronization via Histories and Observables for Monte-Carlo Event Reweighting." SciPost Phys. 18, 054 (2025). arXiv:2410.06342 (2024).

20. Patrick J. Fox, Matheus Hostert, **Tony Menzo**, Maxim Pospelov, Jure Zupan. "Muon-induced baryon number violation." Phys. Rev. D 110, 075015 (2024). arXiv:2407.03450 (2024).

21. Wick Haxton, Kenneth McElvain, **Tony Menzo**, Evan Rule, Jure Zupan. "Effective theory tower for $\mu \to e$ conversion." JHEP 11 (2024) 076. arXiv:2406.13818 (2024).

22. Christian Bierlich, Phil Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Ahmed Youssef, Jure Zupan. "Towards a data-driven model of hadronization using normalizing flows." SciPost Phys. 17, 045 (2024). arXiv:2311.09296 (2023).

23. Christian Bierlich, Philip Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Ahmed Youssef, Jure Zupan. "Reweighting Monte Carlo Predictions and Automated Fragmentation Variations in Pythia 8." SciPost Phys. 16, 134 (2024). arXiv:2308.13459 (2023).

24. Matheus Hostert, **Tony Menzo**, Maxim Pospelov, Jure Zupan. "New physics in multi-electron muon decays." JHEP 10 (2023) 006. arXiv:2306.15631 (2023).

25. Adam Davis, **Tony Menzo**, Ahmed Youssef, Jure Zupan. "Earth mover's distance as a measure of CP violation." JHEP 06 (2023) 098. arXiv:2301.13211 (2023).

26. Reuven Balkin, Eric Madge, **Tony Menzo**, Gilad Perez, Yotam Soreq, Jure Zupan. "On the implications of positive W mass shift." JHEP 05 (2022) 133. arXiv:2204.05992 (2022).

27. Phil Ilten, **Tony Menzo**, Ahmed Youssef, Jure Zupan. "Modeling hadronization using machine learning." SciPost Phys. 14, 027 (2023). arXiv:2203.04983 (2022).

Teaching and Tutorials

Apr. 2026	**Tutorial instructor**, MuonBridge hands-on tutorial, MIAPbP
Jan. 2026	**Tutorial instructor**, (HEP-focused) Agentic programming, CMSDAS, Fermilab
Jun. 2025	**Tutorial instructor**, CTEQ + MCgen summer workshop and training program
2020–2021	**Teaching Assistant**, Physics Department, University of Cincinnati
Summer 2019	**Physics Instructor**, Upward Bound, Columbus State Community College
2015–2019	**Physics and Mathematics Tutor**, The Ohio State University Office of Diversity and Inclusion and Physics Department

Research Software

Agent Frameworks and Toolkits

HEPTAPOD — *Creator and maintainer.* Agentic workflows for symbolic calculations, event generation, and HEP analysis. Related publications: [8, 11].
https://github.com/tonymenzo/heptapod

Orchestral — *Maintainer.* Provider agnostic agentic harness. Related paper: arXiv:2601.02577.
https://orchestral-ai.com

ToolBase — *Co-creator and maintainer.* Packaging, distribution, and deployment of tools and skills for AI agents.
https://github.com/alexr314/toolbase

ToolBench — *Creator and maintainer.* A framework for benchmarking agentic tools and execution environments.
https://github.com/tonymenzo/toolbench

Particle Physics and Computational Methods

MLHAD — Machine learning methods for hadronization modeling and reweighting. Related publications: [12, 14, 16, 18, 19, 22, 27].
https://gitlab.com/uchep/mlhad

RSA — Rejection sampling with automatic differentiation. Related publication: [18].
https://github.com/tonymenzo/RSA

EMD4CPV — Optimal transport methods for detecting CP violation. Related publication: [25].
https://github.com/adamdddave/emd4cpv

MuonBridge — Effective field theory matching and predictions for $\mu \to e$ conversion. Related publication: [21].
https://github.com/Berkeley-Electroweak-Physics/MuonBridge

Educational

MCgen tutorials — Contributions to Monte Carlo event-generation tutorials.
https://gitlab.com/mcgen-ct/tutorials

Seminars and Colloquia

Slides: https://tonymenzo.com/docs/presentations

Sep. 2026	Portable agentic phenomenology *University of Cincinnati HEP/Astro Seminar*
Sep. 2026	Portable agentic phenomenology *Colorado State University HEPPA Seminar* (invited)
Apr. 2026	Agentic phenomenology *BNL HEP Theory Seminar* (invited)
Mar. 2026	Conservation laws and effective (string-)hadronization models *Monash HEP Seminar* (invited)
Jan. 2026	Agents in HEP(-ex) *Fermilab Neutrino Division*
Jan. 2026	Agents in HEP(-ph/-th) *Fermilab Theory Division (chalk talk)*
Oct. 2025	The journey towards differentiable hadronization models *University of Alabama HEP Seminar*
Sep. 2025	The journey towards differentiable hadronization models *Michigan State University HEP Seminar* (invited)
Jul. 2025	Differentiable hadronization models, heavy and light new physics in rare lepton decays, and more... *University of Alabama, Tuscaloosa*
Nov. 2024	Rare μ decays: high-intensity probes of heavy and light new physics *University of Maryland EPT Seminar*
Nov. 2024	Data-driven hadronization models *Rutgers NHETC Seminar*
Nov. 2024	From heavy to light: new physics @ high-intensity μ experiments *Cornell University LEPP Theory Seminar*
Oct. 2024	From heavy to light: new physics @ high-intensity μ experiments *SLAC Theory Seminar*
Oct. 2024	From heavy to light: new physics @ muon factories *UCSD HEP Seminar*
Oct. 2024	From heavy to light: new physics @ muon factories *UCSB HEX-HET Seminar*
May 2024	Building bridges: an effective field theory for muon-to-electron conversion *Berkeley Nuclear Theory Seminar*
Feb. 2024	Optimal transport and CP violation: the Dalitz Mover's Distance *Berkeley 4D Seminar, University of California, Berkeley*
Oct. 2023	Micro from macro (and applications in event generators) *ATLAS Theory Lunch, Lawrence Berkeley National Laboratory*
Sep. 2023	Towards a data-driven model of hadronization *IAIFI Journal Club, MIT*

May 2022 — Modeling hadronization using machine learning
Technion HEP Journal Club

Conference and Workshop Presentations

Slides: https://tonymenzo.com/docs/presentations

Sep. 2026 — Tuning data-driven hadronization models
LPCC MCWG Tuning Forum (invited)

Sep. 2026 — Harness HEPineering
Agentic AI for BSM Event Generation and Analysis Workshop, Fermilab LPC

Sep. 2026 — Benchmarking with HEPTAPOD
Agentic AI for BSM Event Generation and Analysis Workshop, Fermilab LPC

Jun. 2026 — Detecting Ultralight DM via CLFV
CLFV 2026 (invited)

Jun. 2026 — Agentic systems for hep-ph
AI Reasoning in Theoretical Physics, Aspen (chalk talk)

May 2026 — Toward agentic phenomenology
PHENO 2026

Jan. 2026 — AmSC briefing
Online

Dec. 2025 — Agents in HEP
University of Miami Physics Conference

Jun. 2025 — A flavorful and dark cascade for low-scale leptogenesis
NPN 2025

Jun. 2025 — Direct detection of ultralight dark matter with charged-lepton-flavor-violation
FPCP 2025

May 2025 — Road to "differentiable" Pythia? + Half-baked thoughts on energy conservation in hadronization
Pythia Week @ Lund University

May 2025 — Direct detection of ultralight dark matter with charged-lepton-flavor-violation
LULBI @ Weizmann Institute

Apr. 2025 — Direct detection of ultralight dark matter with charged-lepton-flavor-violation
PIKIMO

Sep. 2024 — Signatures of light new physics @ muon factories
NuFact 2024

May 2024 — UV imprints on muon-to-electron conversion
PHENO 2024

May 2024 — Multi-electron muon decays ($\mu \to eeee$)
PIKIMO

May 2023 — Earth Mover's Distance and CP-violation
PHENO 2023

Apr. 2023 — Dark low-scale leptogenesis
PIKIMO, The Ohio State University

Mar. 2023 — Modeling hadronization using machine learning
DIS 2023

Nov. 2022 — Modeling hadronization using machine learning
QCD@LHC 2022

Nov. 2022 — The Wasserstein distance for CP-violation
PIKIMO 13

Oct. 2022	Modeling hadronization using machine learning *AI4EIC*
Jul. 2022	Modeling hadronization using machine learning *RBSM Summer School*